

Dataset Analysis & EDA Report

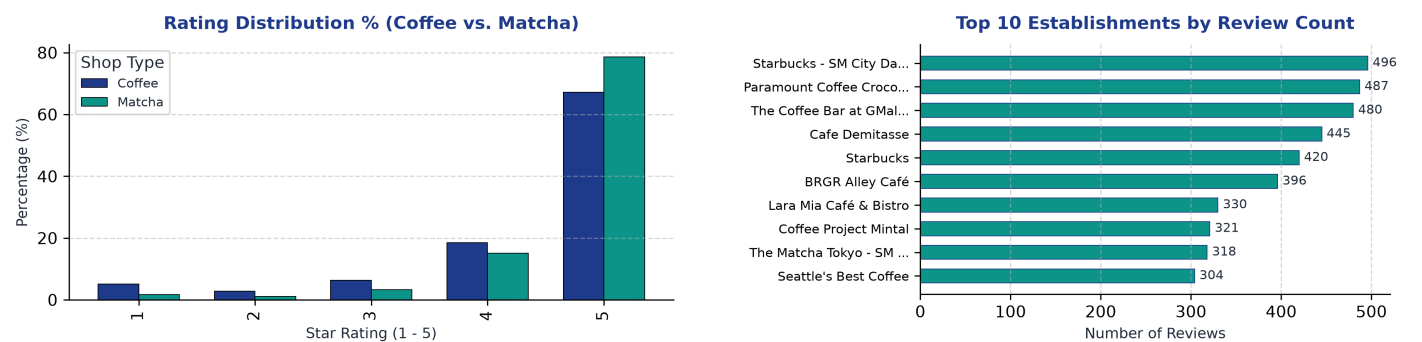
Aspect-Based Sentiment Analysis (ABSA) Dataset Overview & Profiling

This report provides an automated analytical profiling of the Coffeeshops and Matcha Shops master datasets, covering establishment volume balance, rating distributions, linguistic density, emoji frequencies, and n-gram vocabulary metrics.

1. Key Dataset Overview

Metric	Value	Description
Total CSV Files	411	Combined count of Coffeeshop (360) and Matcha (51) files.
Unique Establishments	411	Total unique establishments processed.
Total Raw Rows	20,390	Combined rows across all master CSVs.
Rows with Review Text	13,040 (64.0%)	Reviews containing non-empty text.
Rating-Only Rows (No Text)	7,350 (36.0%)	Empty/missing review text rows.
Duplicate Reviews	7,694	Identical place name and review content matches.

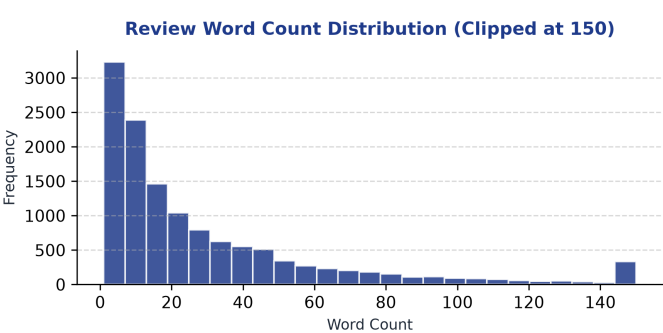
2. Establishment Volume & Rating Visualizations



Volume Concentration & Rating Analysis:

- **Chain Concentration:** Top 3 establishments account for **7.18%** of total reviews, and Top 10 account for **19.60%**.
- **Negative Sentiment Ratio:** 1–3 star ratings account for **14.30%** of Coffee reviews and **6.24%** of Matcha reviews.

3. Textual & Linguistic Profiling



Metric	Word Count
Mean ± Std	30.8 ± 41.8
Median (50%)	16 words
75th Percentile	39 words
90th Percentile	75 words
95th Percentile	108 words
99th Percentile	205 words
Max Outlier	619 words

4. Code-Switching & Emoji Frequency Analysis

Linguistic Profiling Highlights:

- **Code-Switching / Dialect Density:** 3.20% of overall text reviews contain local Bisaya/Tagalog dialect markers (Coffee: 3.27%, Matcha: 2.66%). This validates the requirement for multilingual fine-tuning (e.g. XLM-RoBERTa).
- **Top Emojis Detected (Total 308 unique types):** Top frequencies include #1: 445 occurrences, #2: 271 occurrences, #3: 205 occurrences, #4: 175 occurrences, #5: 170 occurrences, #6: 170 occurrences, #7: 164 occurrences, #8: 145 occurrences, #9: 129 occurrences, #10: 128 occurrences.

5. Vocabulary & N-Gram Analysis

Category	Top Unigrams	Top Bigrams	TTR
Coffee	coffee (4778), place (4333), good (4129), food (3940), not (2301)	good food (470), coffee shop (444), nice place (404), food good (326), good place (283)	0.0342
Matcha	matcha (1018), good (425), place (410), love (217), great (202)	matcha latte (73), best matcha (58), love matcha (54), good matcha (50), must try (43)	0.1030
Overall	Combined corpus analysis	Combined corpus analysis	0.0327